GAURAV SHINDE

AI ENGINEER | GENERATIVE AI | RAG | LLM FINE-TUNING | AI AGENTS
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PROFESSIONAL SUMMARY

AI Engineer at TCS with hands-on experience building Generative AI, Retrieval-Augmented Generation (RAG), LLM fine-tuning, and AI agent systems using open-source LLMs (Llama), LangChain, LangGraph, vector databases, and FastAPI. Deployed a RAG chatbot for a public-sector bank client. Fine-tuned small language models with LoRA/QLoRA and built evaluation-driven pipelines covering retrieval quality, hallucination detection, and leakage-controlled benchmarks. Targeting AI Engineer and ML Engineer roles.

TECHNICAL SKILLS

Generative AI & LLMs: Large Language Models (LLMs), Llama 3.1/3.2, Small Language Models (SLMs), Fine-Tuning, SFT, PEFT, LoRA/QLoRA, 4-bit Quantization, Unsloth, Hugging Face Transformers, TRL, GGUF, llama.cpp, Ollama, Groq API, Prompt Engineering

RAG & Retrieval: Retrieval-Augmented Generation (RAG), Vector Databases, ChromaDB, Embeddings, Sentence Transformers, BGE, Semantic Search, Document Chunking, Document Ingestion, Citation-Grounded Generation, Retrieval Evaluation

AI Agents: LangChain, LangGraph, Agentic AI, Stateful Workflows, Conditional Routing, Tool Calling, Self-Correction Loops, Retrieval-Based Memory, Text-to-SQL

Evaluation & Safety: LLM Evaluation, RAGAS, Semantic Similarity, NDCG, Execution Accuracy, Hallucination Detection, Regression Testing, Error Analysis, Leakage Prevention, Threat Modeling, Input Validation, SQL Allowlisting

Engineering & MLOps: Python, PyTorch, FastAPI, REST APIs, Streamlit, Docker, Docker Compose, GitHub Actions, CI/CD, pytest, Git, Model Serving

Data: SQL, SQLite, Pandas, NumPy, Embedding Pipelines

PROFESSIONAL EXPERIENCE

AI Engineer, Tata Consultancy Services (TCS)Navi Mumbai | 2024 – Present

- Built and deployed a client-facing RAG chatbot enabling branch staff of a public-sector bank to resolve recurring transaction issues, including interest-rate errors and data-correction requests, in real time.
- Architected the RAG pipeline using LangChain, Llama 3.1 8B with Ollama, BAAI/bge-small-en-v1.5 embeddings, and Oracle Database as the vector store for semantic retrieval and LLM response generation.
- Analyzed recurring service-desk issues and designed an AI-assisted resolution workflow that enabled branches to resolve 75% of identified requests without backend analyst escalation.

PROJECTS

SQLMind – Self-Correcting Text-to-SQL AI Agentgithub.com/Gaurav-shinde2207/SQLmind
LangGraph, Groq, SQLite, FastAPI, Sentence Transformers

- Built a LangGraph agent for natural-language-to-SQL generation with intent parsing, example retrieval, SQL generation, execution, validation, and up to 3 bounded self-correction retries; exposed through a CLI and FastAPI.
- Implemented secure read-only SQL execution with a SELECT/WITH-only statement gate, read-only database connection, LIMIT guard, and query timeout; wrote a STRIDE threat model covering cache poisoning.
- Evaluated three configurations on 28 held-out Chinook questions with leakage-controlled splitting: baseline 60.7%, retry-only 57.1%, retry + correction memory 57.1% execution accuracy. Memory gave no measurable gain (overlapping 95% CIs); nested queries were weakest at 28.6%. Added 150 offline tests.

SQL2NL – Fine-Tuned Small Language Model (SLM) for SQL-to-Business-Languagegithub.com/Gaurav-shinde2207/SQL2NL
Llama 3.2 3B, Unsloth, QLoRA, TRL, llama.cpp, FastAPI, Ollama, Hugging Face Hub

- Fine-tuned Llama 3.2 3B Instruct with QLoRA on 1,500 SQL-to-explanation examples covering CTEs, window functions, and JOINs; trained in under 1 GPU-hour on a Kaggle T4.
- Exported a ~2 GB Q4_K_M GGUF model, published it on Hugging Face (huggingface.co/ShindeGaurav-2207/SQL2NL), and served it through FastAPI with llama.cpp/Ollama; added pytest tests and GitHub Actions CI.
- Built an evaluation harness using keyword coverage, MiniLM semantic similarity, and latency; 6/8 held-out queries passed, with 0.774 average semantic similarity and 65.1% keyword coverage.

FinSight – Financial Document Q&A with RAGgithub.com/Gaurav-shinde2207/finsight-financial-rag-system
FastAPI, Streamlit, ChromaDB, BGE, PyMuPDF, Groq (Llama 3.1 8B), RAGAS, Docker Compose

- Built an end-to-end RAG system with PDF upload, page-level extraction, section-aware overlapping chunks, BGE embeddings, persistent ChromaDB vector storage, and grounded answer generation through REST APIs.
- Attached file/page/section citations and similarity scores to answers; returned an insufficient-context response instead of calling the LLM when no relevant chunks were retrieved.
- Built evaluation with RAGAS metrics, NDCG@k, deterministic sentence-level hallucination detection, and a benchmark runner that flags faithfulness regressions; exposed through a FastAPI /evaluate endpoint.

EDUCATION & CERTIFICATION

B.Tech, Mechanical Engineering | Annasaheb Dange College of Engineering and Technology, Ashta2019 – 2023

Certification: Generative AI with Large Language Models